

Intro to Apache Spark

What is Spark?

- ▶ Apache SparkTM is a multi-language engine for executing data engineering, data science, and machine learning on single-node machines or clusters.
 - ▶ Written in Scala. Supports Scala, Java, Python, and R.
- ▶ It is streamlined for fast real time as well as batch performance.
- ▶ It can be used in a Spark cluster and coexists with a HDFS cluster using HDFS for input and output.
- ▶ The MapReduce concept is useful in understanding how Spark functions work!

Spark Installation and Setup

- ▶ Download latest version from <http://spark.apache.org/downloads.html> by selecting the package type of "Pre-built for Hadoop 3 and later." The download is a compressed tarball, called [spark-4.2.0-bin-hadoop3.tgz](#)
- ▶ Create a top-level folder and unpack inside it as follows:

```
cd ~  
mkdir spark-install  
cd spark-install  
tar xzvf ../Downloads/spark-4.2.0-bin-hadoop3.tgz  
ln -s spark-4.2.0-bin-hadoop3 spark
```

- ▶ Then add Spark programs to your path as follows (all in one line):

```
echo "export PATH=~/.spark-install/spark/bin:~/.  
      spark-install/spark/sbin/:$PATH" >> ~/.bashrc  
source ~/.bashrc
```
- ▶ Test it by starting one of the shells. For example, try [spark-shell](#) (for Scala-based shell) or try [pyspark](#) for Python-based shell. Use [Ctrl-d](#) ([Ctrl-z](#) on MS Windows) to terminate the shell.

Interactive Data Analysis(1)

- ▶ Start the interactive spark shell:

```
spark-shell
```

- ▶ Let's make a new Dataset from the text of an input file:

```
scala> val textFile = spark.read.textFile("word-count/input/  
    Alice-in-Wonderland.txt")  
textFile: org.apache.spark.sql.Dataset[String] = [value:  
    string]
```

- ▶ We can get values directly from the Dataset, such as:

```
scala> textFile.count()  
res0: Long = 3840
```

```
scala> textFile.first()  
res1: String = ***This is the Project Gutenberg Etext of  
    Alice in Wonderland***
```

```
scala> textFile.tail(4)  
res2: Array[String] = Array(remembering her own child-life,  
    and the happy summer days., "", "  
    THE END", ?)
```

Interactive Data Analysis(1)

- ▶ Start the interactive spark shell in Python:

```
pyspark
```

- ▶ Let's make a new Dataset from the text of an input file:

```
>>> textFile = spark.read.text("input/Alice-in-Wonderland.txt")
```

- ▶ We can get values directly from the Dataset, such as:

```
>>> textFile.count()
3840
```

```
>>> strings.first()
Row(value='***This is the Project Gutenberg Etext of Alice
      in Wonderland***')
```

```
>>> strings.tail(4)
[Row(value='remembering her own child-life, and the happy
      summer days. '), Row(value=''), Row(value='
      THE END'), Row(value='\x1a')]
```

Spark Concepts

- ▶ Spark is a distributed computing framework. It can run on a single machine or on a cluster of machines. It can be run in a standalone mode or on top of Hadoop YARN
- ▶ Spark Application, SparkSession.
- ▶ Jobs, Stages, Tasks
- ▶ Transformations and Actions. Lazy Evaluation.
 - ▶ Examples of Transformations: `map`, `filter`, `orderBy`, `groupBy`, `select`, `join`, etc
 - ▶ Examples of Actions: `count`, `collect`, `take`, `show`, `saveAsTextFile`, etc
- ▶ Transformations can be narrow or wide. Narrow transformations can be executed on a single node, while wide transformations require data to be shuffled across nodes.

Interactive Data Analysis (2)

- Now, let's use a transformation on the Dataset:

```
>>> linesWithAlice = strings.filter(strings.value.contains("
    Alice"))
>>> linesWithAlice.count()
393
```

Interactive Data Analysis (3)

- ▶ We can pull data into a cluster-wide in-memory cache. This will speed up performance for a dataset that is being used repeatedly.

```
>>> linesWithAlice.cache()
```

```
>>> linesWithAlice.count()  
393
```

- ▶ These same functions can be used on very large data sets, even when they are striped across tens or hundreds of nodes. We can also do this interactively by connecting bin/spark-shell to a cluster, as we will see later.
- ▶ Note that a Dataset by itself isn't parallel/distributed by default. We will see later how to work with distributed data.

First Standalone Example (Java)

- ▶ Examine the first simple app example under CS435-435-535-resources/examples/Spark in the **simpleapp** folder.
- ▶ Examine the code directly here: **SimpleApp.java**
- ▶ We can export the jar file from Eclipse (or other IDEs). Create a standard Java project and include all Spark jar files from `~/spark-install/spark/jars` as *External JARs* for the project.
- ▶ Now, we submit the jar file to run on Spark as follows:

```
spark-submit --class "SimpleApp" --master local[4]  
simpleapp.jar
```

- ▶ Note that we are running Spark locally on our system. That is denoted by the `local[4]` option. The number in square brackets it is the number of threads to use.
- ▶ To redirect the verbose messages from Spark, redirect as follows:

```
spark-submit --class "SimpleApp" --master local[4]  
simpleapp.jar 2> log
```

- ▶ The `spark-submit` app has many options. Just run it without any arguments to get the full list of options. Note the order of application of options from highest to lowest is as follow:
 - ▶ Explicit settings in the application.
 - ▶ Commandline arguments to `spark-submit`.
 - ▶ Options from Spark configuration files from the installation.

First Standalone Example (Python)

- ▶ Examine the first Python example in the CS435-535-resources/examples/Spark in the `simpleapp-python` folder.
- ▶ Examine the code directly here: `SimpleApp.py`
- ▶ Now, we run the python script on Spark as follows:

```
spark-submit --master local[4] SimpleApp.py
```
- ▶ Note that we are running Spark locally on our system. That is denoted by the `local[4]` option. The number in square brackets it is the number of threads to use.
- ▶ To redirect the verbose messages from Spark, redirect as follows:

```
spark-submit --master local[4] SimpleApp.py 2> log
```

Installing Spark+Python ecosystem

- ▶ Here is a high level overview of the installation steps
 - ▶ Install Python
 - ▶ Install Java JDK
 - ▶ Install Jupyter Notebook
 - ▶ Install Apache Spark
 - ▶ Download the spark tgz file and unpack it (similar to Hadoop)
 - ▶ Set the paths for your login so that the spark commands can be found from the command line.

Interactive Spark with Python

- ▶ Simply start the Python Spark shell with the the command `pyspark`.
- ▶ Using Pyspark from Jupyter lab is the better way to go. Open the `simpleapp-jupyter` example in Jupyter and follow along.

Operation of a Spark Application

- ▶ A Spark application consists of a **driver** program that launches various parallel operations on the cluster. It creates distributed datasets on the cluster, and then applies operations to them.
- ▶ Driver programs access Spark via a **SparkContext** object, which represents a connection to a computing cluster. We use this to create **Resilient Distributed Datasets (RDDs)** on which we can then apply various parallel operations.
- ▶ The driver program manages a number of nodes called executors. These are used to parallelize the operations.
- ▶ Spark programs can be written via interactive shells (for Scala and Python, for example) or they can be standalone (for Scala, Python, Java etc). For production environments, standalone would be the norm.

Spark Data Representation

- ▶ Spark supports three data representations: **Resilient Distributed Dataset (RDD)**, **Dataframe**, **Dataset**.
- ▶ A *RDD* is a collection of objects that stores data partitioned across the nodes of the cluster, enabling parallel processing. This is the fundamental underlying data structure in Spark.
- ▶ A *Dataframe* is also a distributed collection organized into named columns. Dataframe are recommended because of ease of usage.
- ▶ A *Dataset* is an extension of *Dataframe* with benefits of both *RDD* and *Dataframe*. Supported in Scala and Java but not supported in Python (as it needs a statically typed language)